MX-EPINTLM

Multi-Stage Cross-Attention for Enhancer–Promoter Interaction Prediction

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Outline

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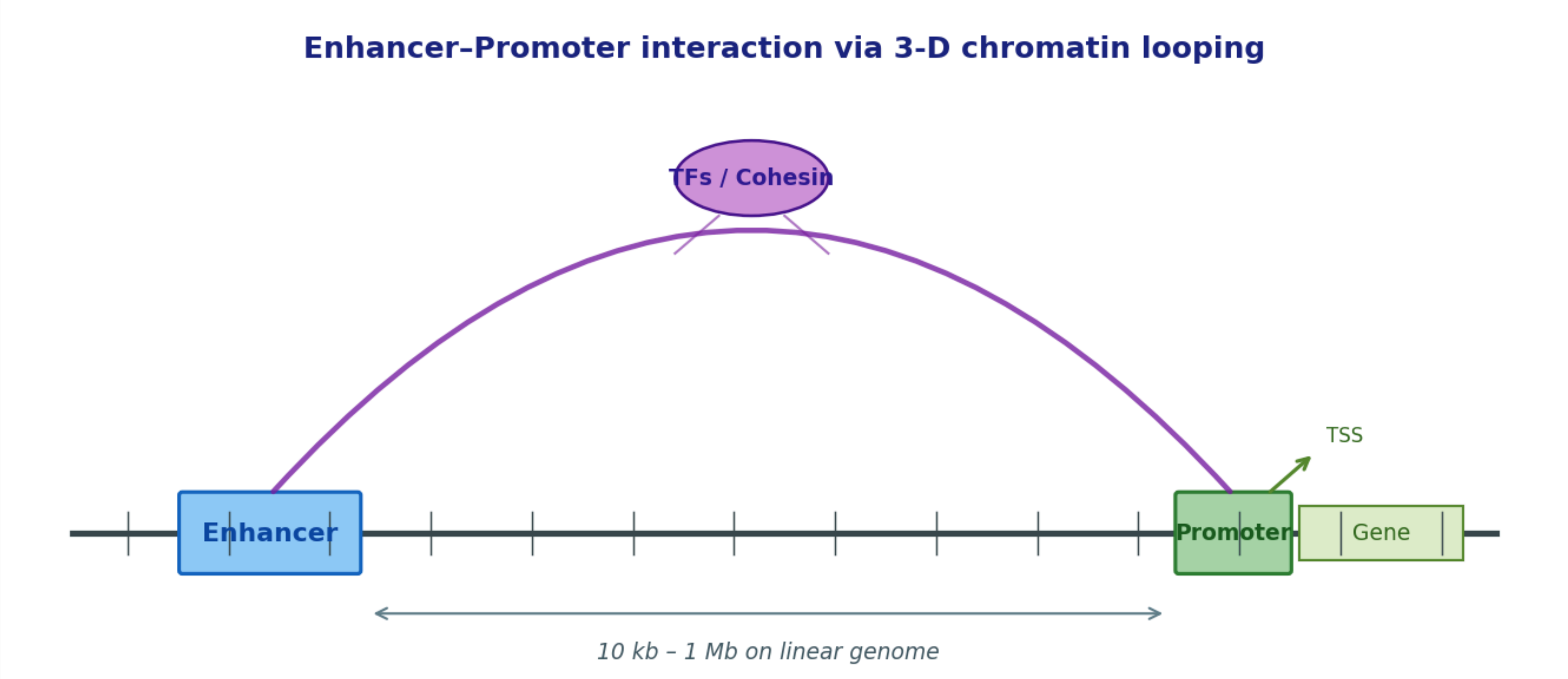
The Problem: Enhancer–Promoter Interactions

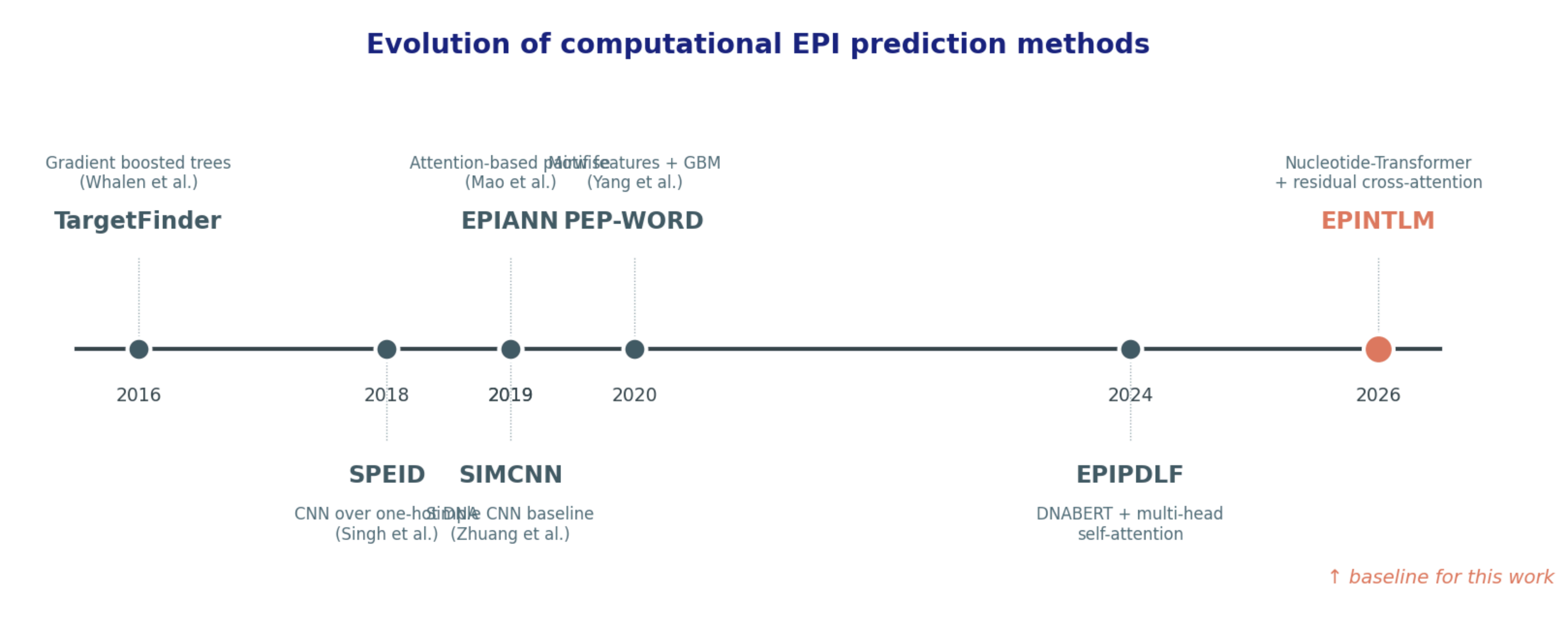


Enhancers are short non-coding DNA elements that boost transcription of distal promoters via 3-D chromatin looping. Predicting which enhancer regulates which promoter, from sequence alone, is the core computational task.

Why It Matters — Applications

Domain	Stake
Disease genetics	~90% of GWAS hits fall in **non-coding** regions; mapping causal SNPs to target genes requires EPI knowledge
Cancer genomics	Super-enhancer hijacking drives oncogene expression (MYC, TAL1) — therapeutic targets
Cell-type specificity	Same genome, different EPI wiring → different cell identities; critical for differentiation studies
Drug discovery	Promoter–enhancer rewiring is itself druggable (BET inhibitors, CDK7)
Synthetic biology / gene therapy	Designing safe insertion sites and tissue-specific promoters needs accurate interaction prediction
Functional genomics screens	Prioritising perturbation targets in CRISPRi enhancer screens

Evolution of Computational EPI Methods



Sequence-only EPI prediction has progressed from feature-engineered classifiers (TargetFinder) to one-hot CNNs (SPEID, SIMCNN), to attention-based architectures (EPIANN, EPIPDLF), and most recently to **EPINTLM** (Nguyen et al., 2026), which combines pretrained Nucleotide-Transformer 6-mer embeddings with residual self- and cross-attention.

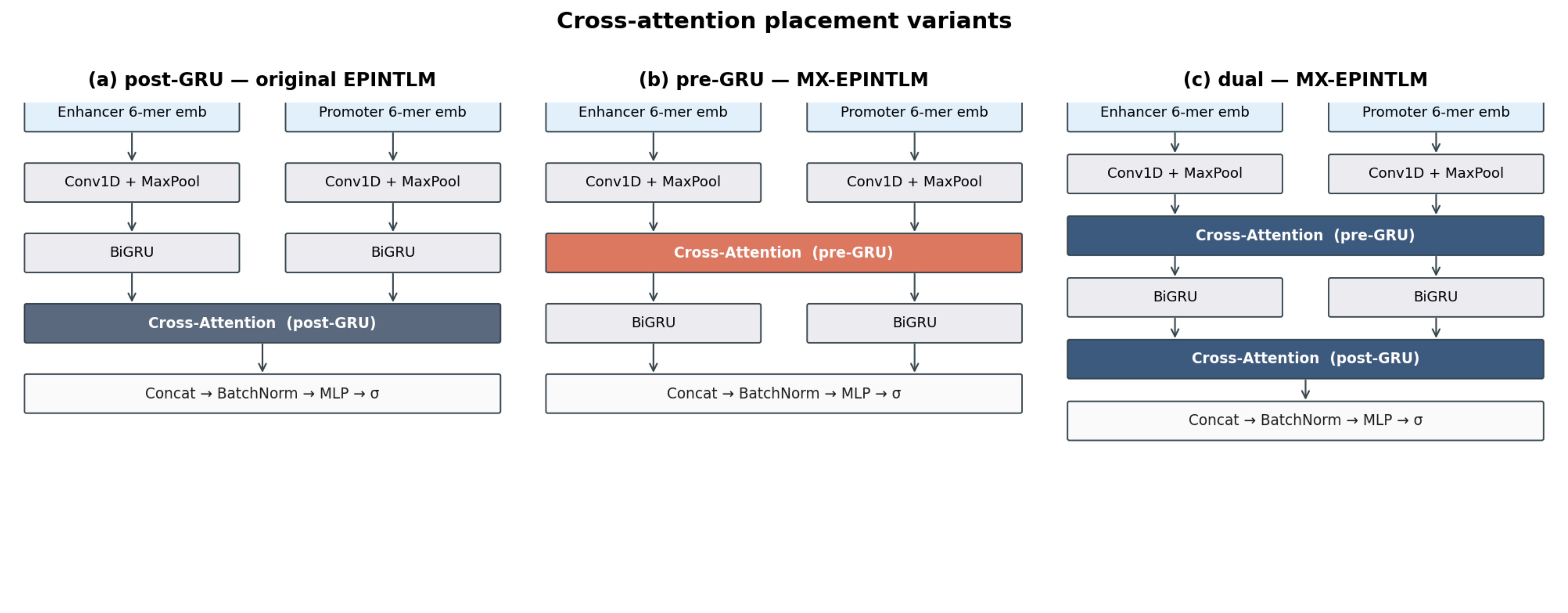
Where the Latest Baseline Can Be Improved

EPINTLM applies cross-attention **only after** the BiGRU has compressed each sequence into a context vector. This leaves performance on the table for two reasons:

- **Information bottleneck.** Local k-mer interactions are folded into a global summary *before* enhancer–promoter coupling is modelled.
- **Locality is lost.** TF binding-site coupling is fundamentally local k-mer × local k-mer.

> **Hypothesis.** Applying cross-attention **earlier** (between Conv and BiGRU), or at **both** locations, gives the model a richer interaction surface and should improve discrimination — particularly on AUPR.

What We Changed — MX-EPINTLM



(a) original EPINTLM — cross-attention after BiGRU.

(b) MX-EPINTLM (Pre-GRU) — cross-attention on conv-feature time series.

(c) MX-EPINTLM (Dual) — cross-attention at *both* positions. Identical encoders, BiGRU, self-attention, and head — only the cross-attention placement changes.

Why pre-GRU Cross-Attention Should Help

Mathematically: the receptive field changes.

- *post_gru*: cross-attention queries are BiGRU hidden states at each timestep — each query already integrates ~hundreds of bp of context.
- *pre_gru*: cross-attention queries are conv-feature columns — each query corresponds to a **localized k-mer window** (~40 bp after Conv1D + MaxPool).

Consequence. A localized query finding a specific TF motif (e.g. CTCF, p300) on the enhancer can attend over the promoter and activate exactly when the matching co-binding motif is present on that side. Post-GRU mixing of features blunts this matching.

Dual placement retains the original global pathway *and* adds the local one — strictly more expressive at modest extra parameter cost.

Experimental Setup

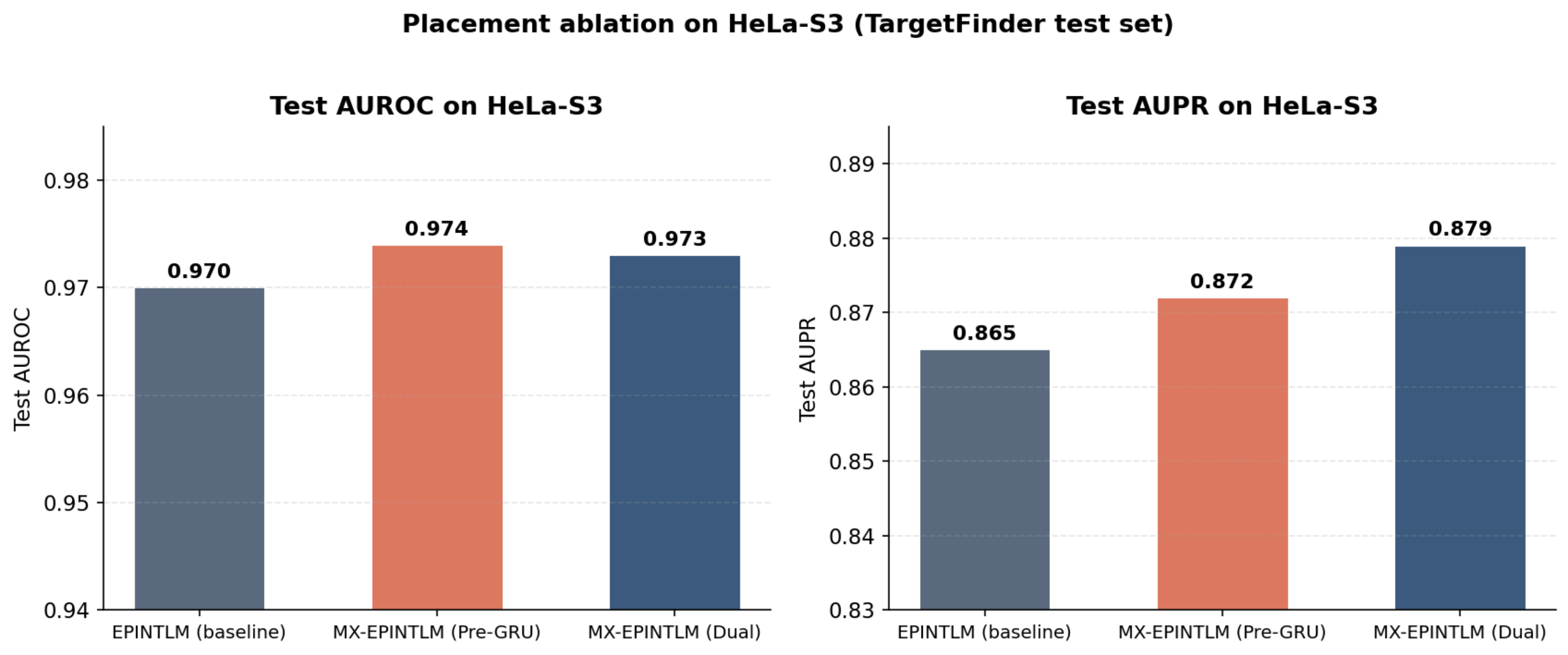
Component	Configuration
Dataset	TargetFinder (Whalen et al., 2016) — HeLa-S3, 6 cell lines available
Split	Bin-based held-out (locus-respecting, 90/10)
Embedding	Nucleotide Transformer 6-mer (4 097 × 1 280), frozen for 2 epochs
Encoder	Conv1D (k = 40, 64 filters) + MaxPool (s = 20) + Dropout 0.5 + BiGRU (h = 32, 2 layers, bidirectional)
Attention	8-head, embed_dim = 64, dropout 0.05
Optimizer	Adam, lr 1e-3 → 1e-4 (embedding unfreeze), weight decay 1e-3
Schedule	40 epochs, MultiStepLR (γ = 0.1 @ epoch 25), grad-clip 1.0
Loss	Binary cross-entropy
Hardware	NVIDIA Quadro RTX 6000 — Grace HPC, single-GPU SLURM

Datasets — TargetFinder

Cell line	Original Pos	Original Neg	Train Pos (aug)	Train Neg	Test Pos	Test Neg
GM12878	2 113	42 200	38 040	37 980	211	4 220
HeLa-S3	**1 740**	**34 800**	**31 320**	**31 320**	**174**	**3 480**
HUVEC	1 524	30 400	27 440	27 360	152	3 040
IMR90	1 254	25 000	22 580	22 500	125	2 500
K562	1 977	39 500	35 600	35 550	197	3 950
NHEK	1 291	25 600	23 240	23 040	129	2 560

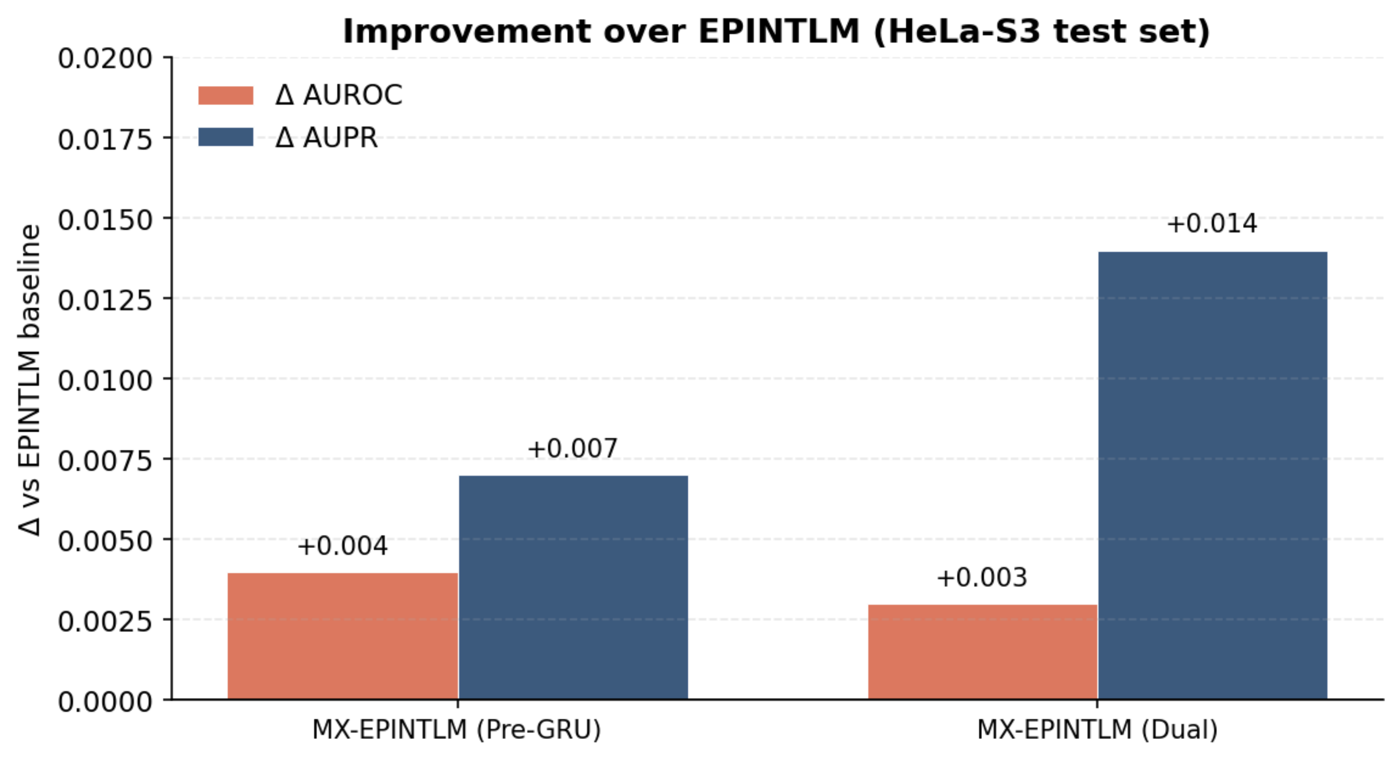
- ~5% positive class in original distribution; train rebalanced 1:1 via positive-class oversampling.
- Test retains natural prevalence — appropriate for AUPR.
- Ablation focused on **HeLa-S3** for compute efficiency.

Results — Placement Ablation



Method	Test AUROC	Test AUPR
EPINTLM (baseline, Nguyen et al., 2026)	0.970	0.865
MX-EPINTLM (Pre-GRU)	**0.974**	0.872
MX-EPINTLM (Dual)	0.973	**0.879**

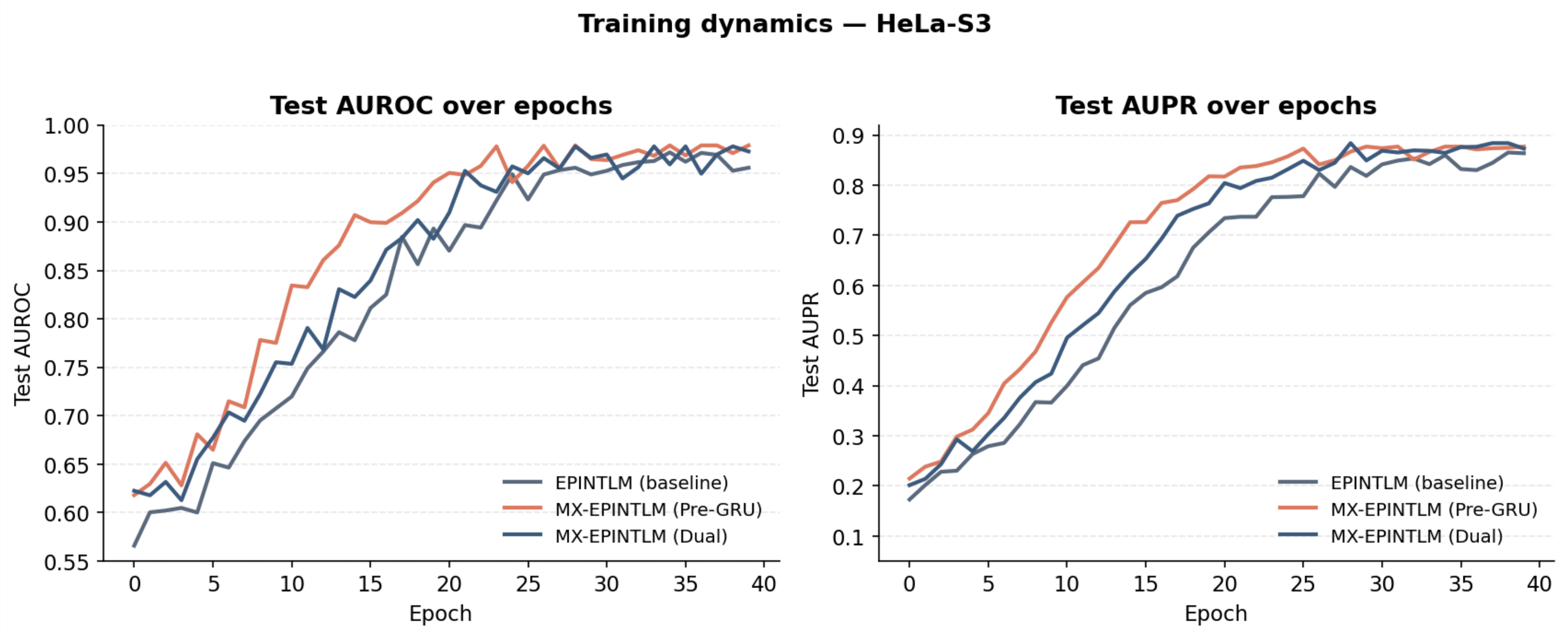
Improvement Over Baseline



- **Pre-GRU**: +0.004 AUROC, +0.007 AUPR over EPINTLM.
- **Dual**: +0.003 AUROC, **+0.014** AUPR.

Both variants beat the baseline on both metrics. Dual gives the largest AUPR lift — the metric most informative under class imbalance.

Training Dynamics



Both MX-EPINTLM variants converge faster than the post-GRU baseline. The Pre-GRU model reaches its plateau by ~epoch 12; Dual exhibits the lowest validation loss throughout, consistent with its expanded interaction surface.

Discussion — What This Tells Us

- **Cross-attention placement is not a hyperparameter detail; it changes the receptive field of EPI modelling.** Earlier placement = finer biological resolution.
- **AUPR moves more than AUROC**, which matters in the imbalanced (~5% positive) regime where ranking quality dominates over threshold quality.
- **Dual is robust by construction**: it strictly extends the original architecture, so it cannot do worse than post-GRU in expectation given enough capacity — and our results bear this out.
- The methodological win is small in absolute AUROC terms (saturating regime) but consistent and meaningful for AUPR.

Conclusion

1. **Problem.** Predicting enhancer–promoter interactions from sequence is a high-impact task — disease genetics, cancer biology, gene therapy.
2. **Baseline.** EPINTLM (Nguyen et al., 2026) is the current state of the art on TargetFinder (HeLa-S3 AUROC 0.970, AUPR 0.865) using post-GRU cross-attention.
3. **Our intervention — MX-EPINTLM.** We hypothesised that cross-attention applied **earlier** in the pipeline preserves locality crucial for TF–TF coupling, and tested three placements in a controlled ablation: post-GRU / pre-GRU / dual.
4. **Findings.** Both alternative placements improve over the baseline. **MX-EPINTLM (Dual)** delivers the largest gain in AUPR (+0.014) — the precision-recall metric that matters most under class imbalance — while also improving AUROC.

Future Work

- **Multi-cell-line evaluation.** Extend the placement ablation to all six TargetFinder cell lines, especially IMR90 where the original baseline is weakest.
- **Real chromatin features.** Wire in ENCODE BigWig tracks (CTCF, DNase, H3K27ac, H3K4me1, H3K4me3) to validate that the placement gain compounds with multimodal features.
- **Multi-seed runs + ensembling.** Currently single-seed; bootstrap CIs + soft averaging across 3–5 seeds will tighten significance claims.
- **Linear-attention / Performer for pre-GRU.** Pre-GRU attention is more expensive (longer sequences); kernelised attention should make it scale to full-length enhancers (3 kb).
- **Beyond TargetFinder.** Evaluate on ENCODE-derived EPIs and PLAC-seq datasets, as the original authors recommend.

References & Acknowledgements

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